

paper001

Eoin Murray¹ and Timothy O'Leary¹

*For correspondence:

¹Department of Engineering, University of Cambridge

Abstract To be drafted.

Introduction

To be drafted.

Results

To be drafted.

Discussion

To be drafted.

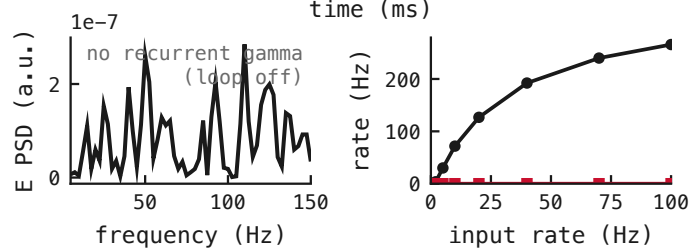
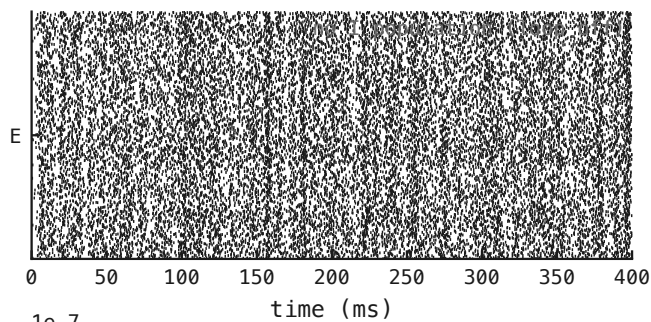
Materials and methods

To be drafted

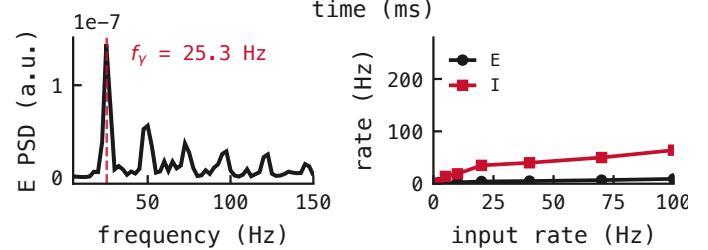
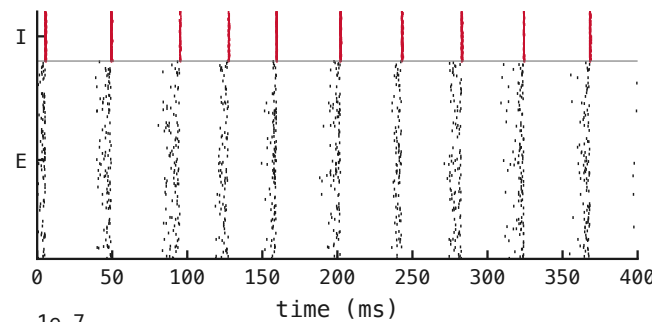
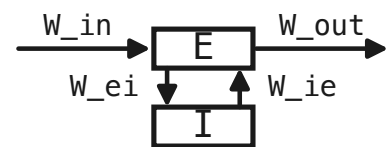
Figures

References

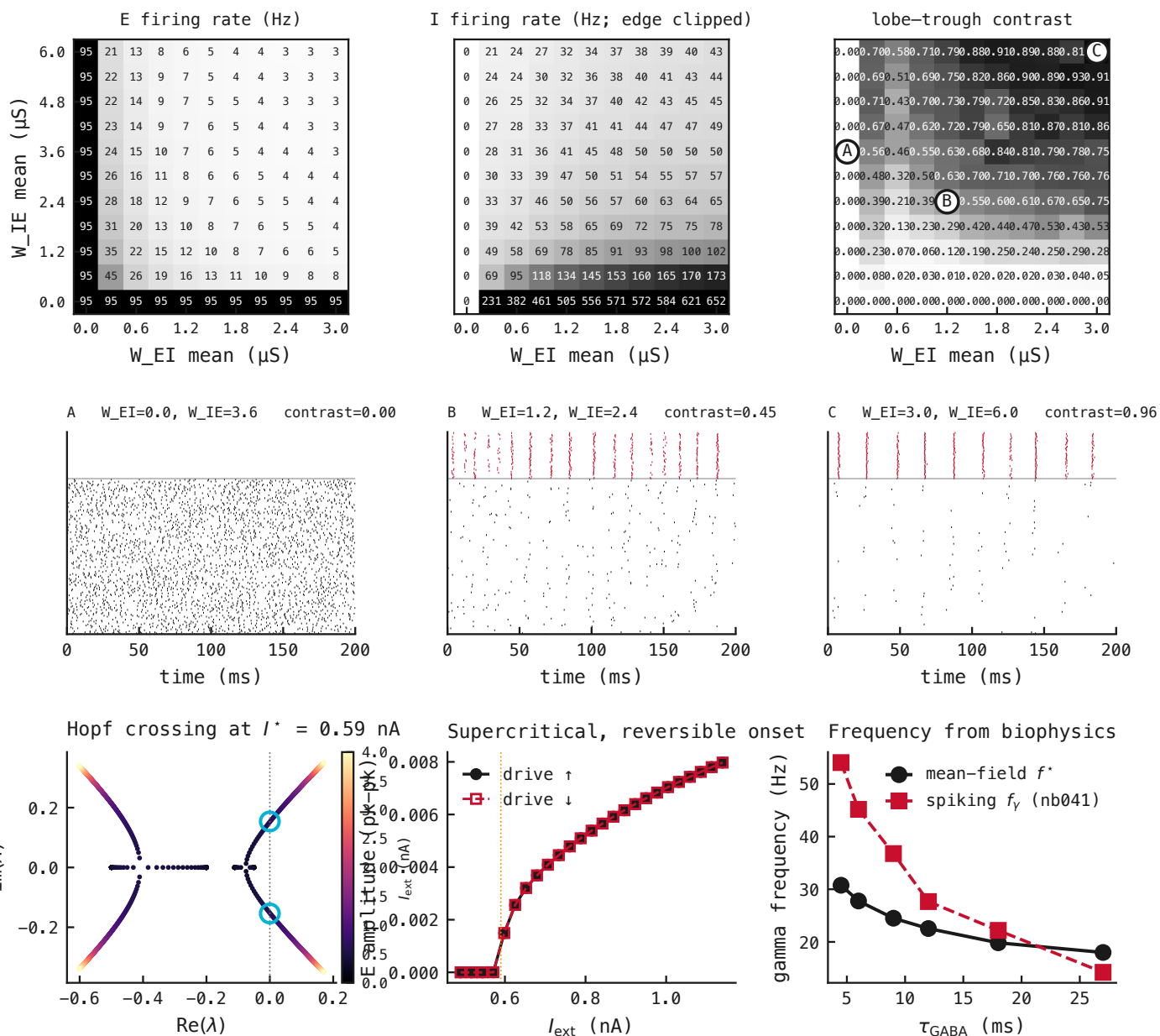
A COBA – recurrent loop off



B PING – recurrent loop active

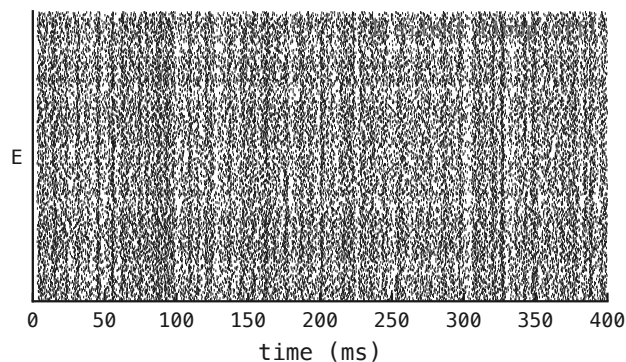


Appendix 0 Figure 1

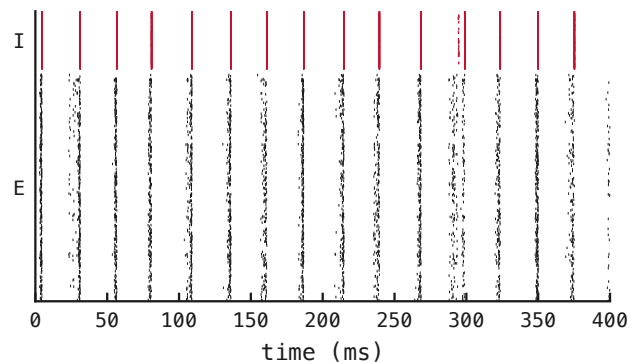


Appendix 0 Figure 2

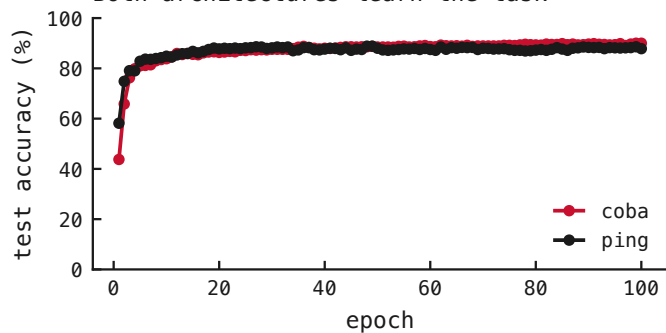
COBA – loop off



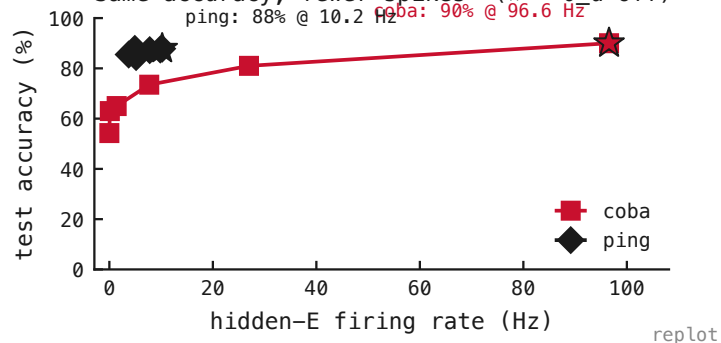
PING – loop on



Both architectures learn the task

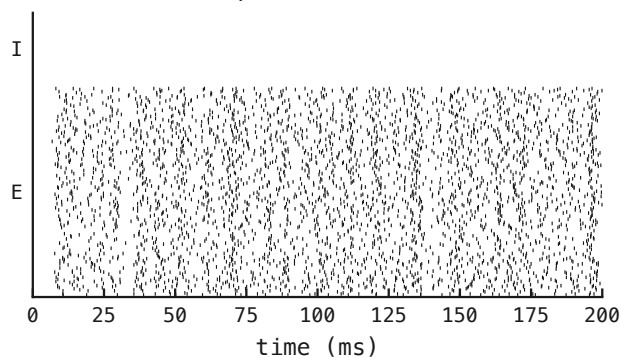


Same accuracy, fewer spikes ($\star = \theta_u$ off)

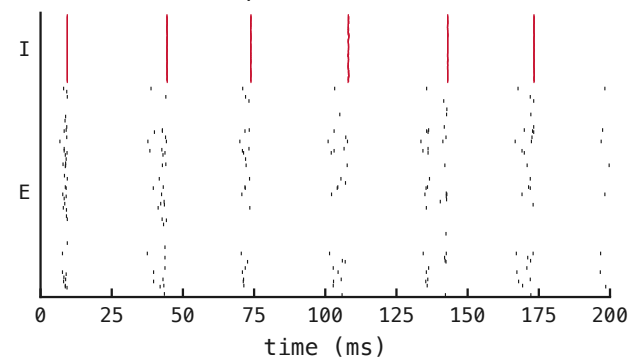


Appendix 0 Figure 3

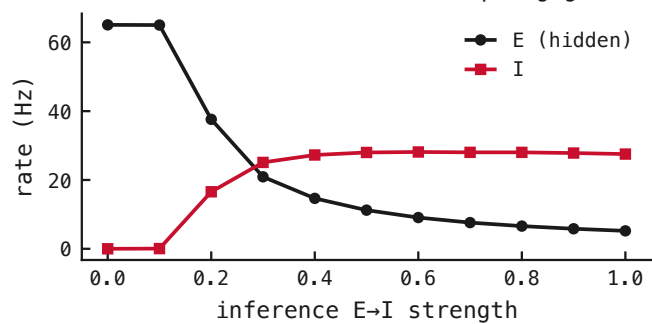
ei = 0 – loop off (COBA)



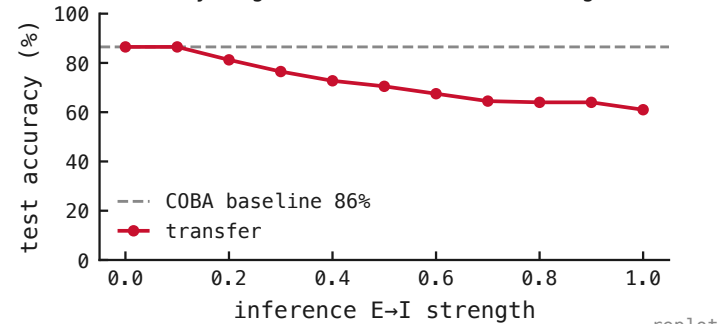
ei = 1 – loop on (PING)



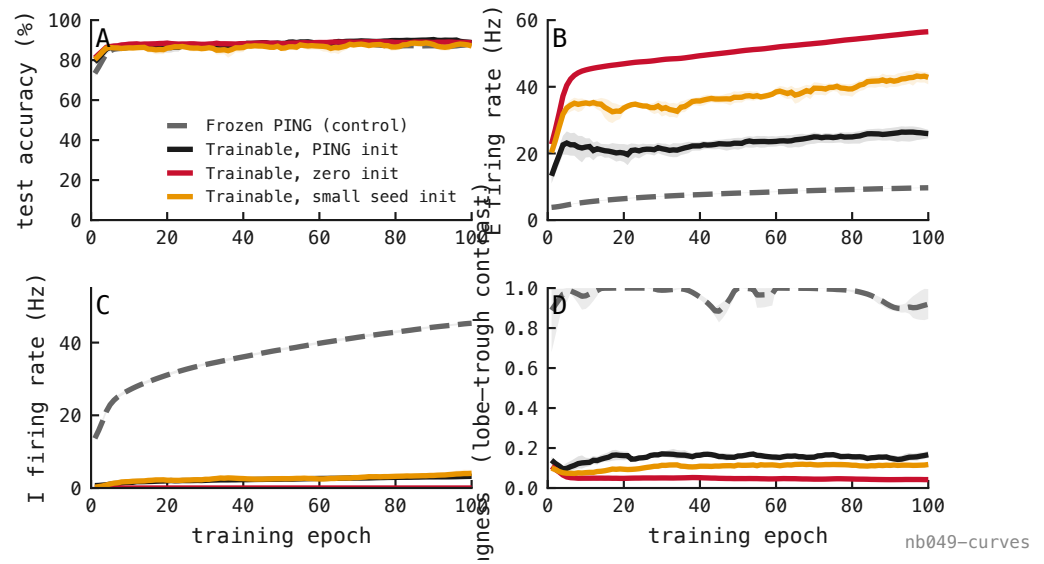
E rate falls $\approx 10\times$ as the loop engages



Accuracy degrades without retraining

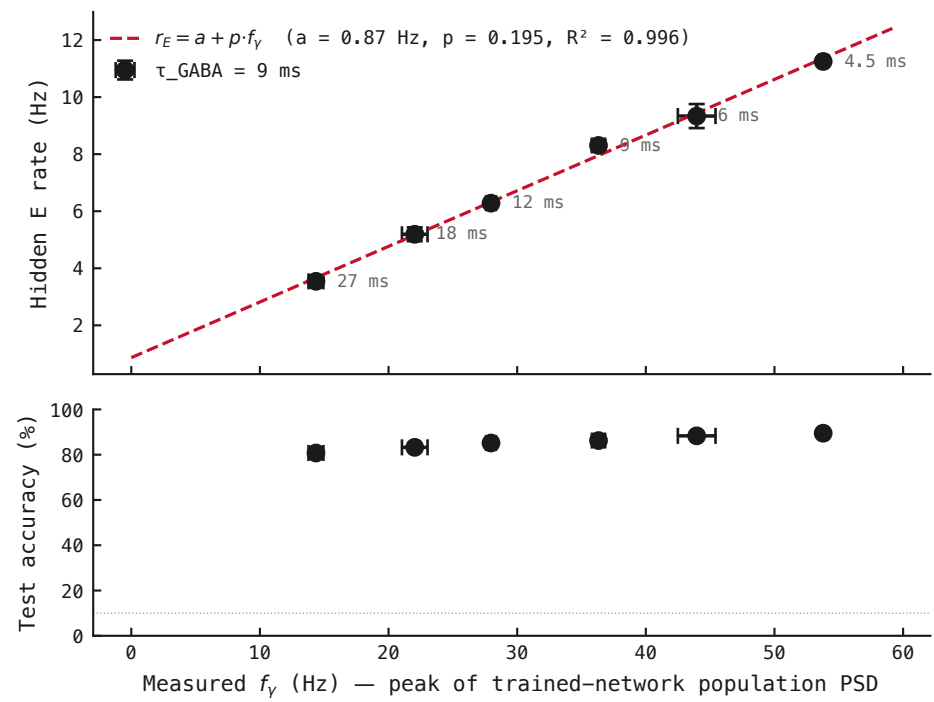


Appendix 0 Figure 4

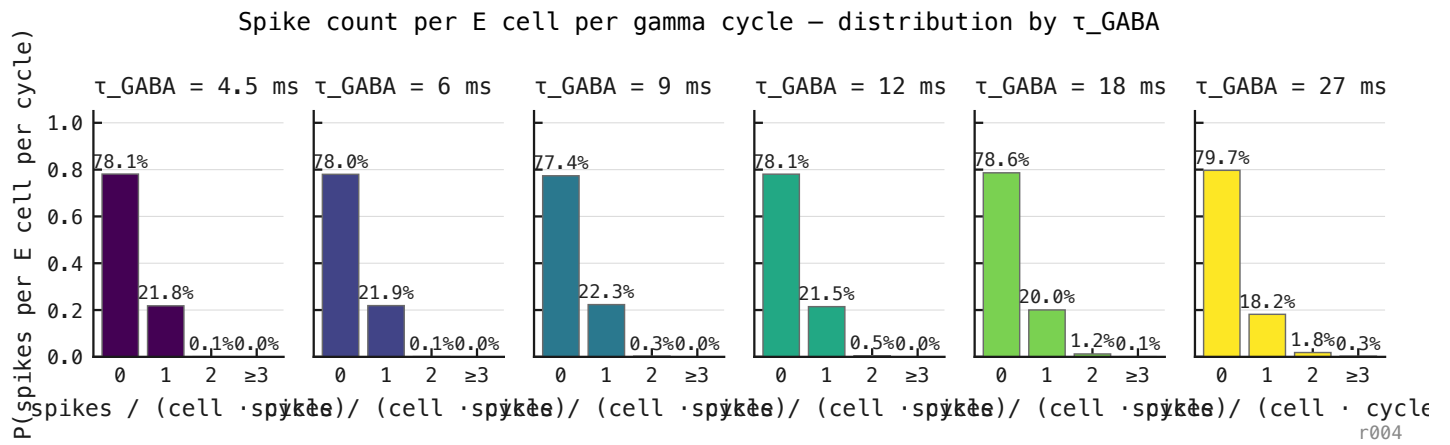


Appendix 0 Figure 5

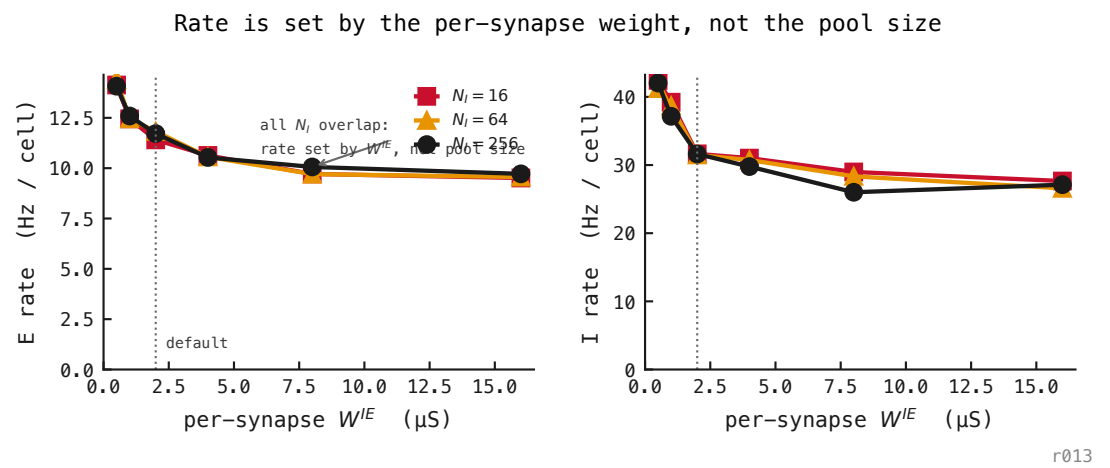
Retrained τ_{GABA} sweep — post-training E rate tracks measured f_Y



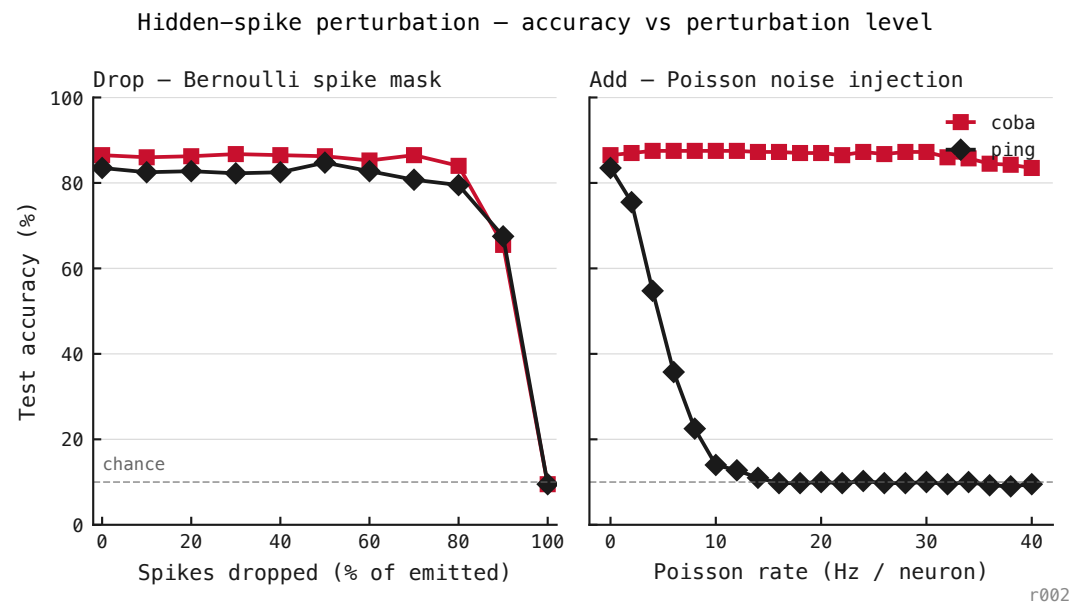
Appendix 0 Figure 6



Appendix 0 Figure 7

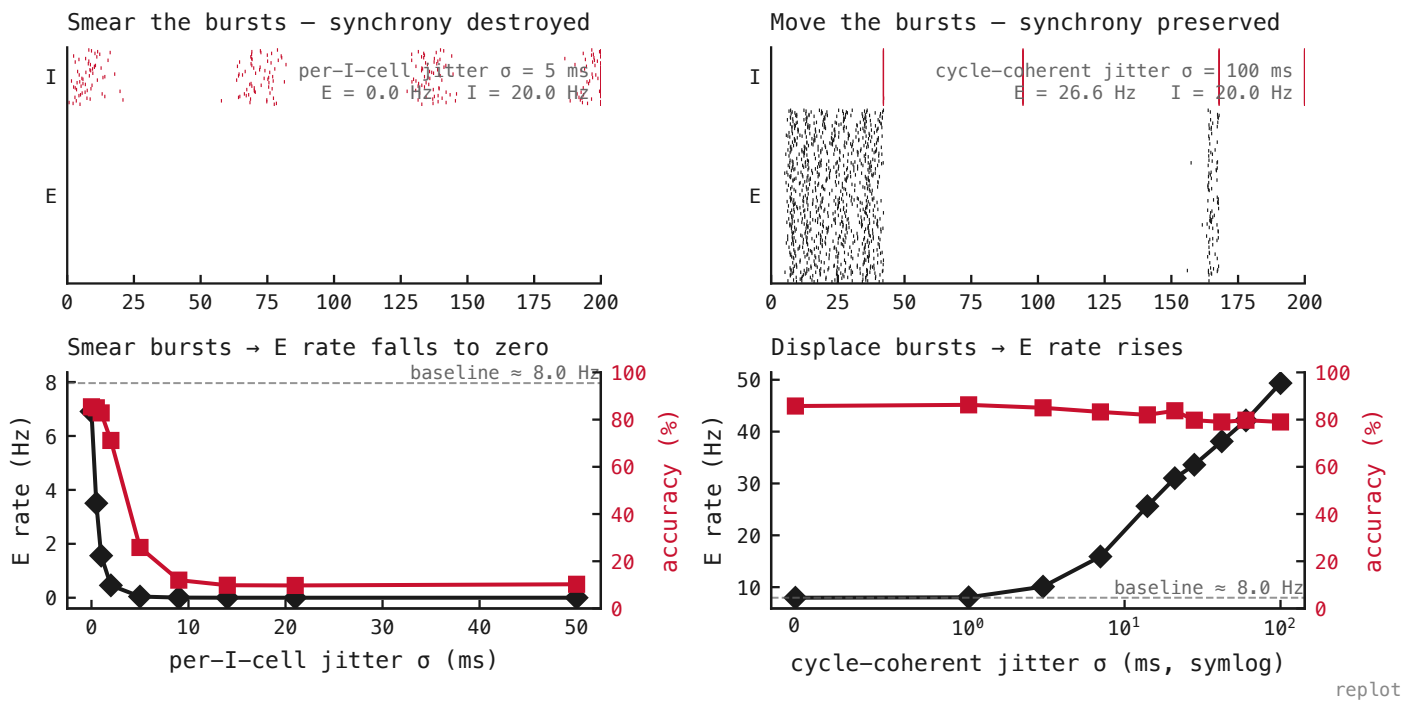


Appendix 0 Figure 8



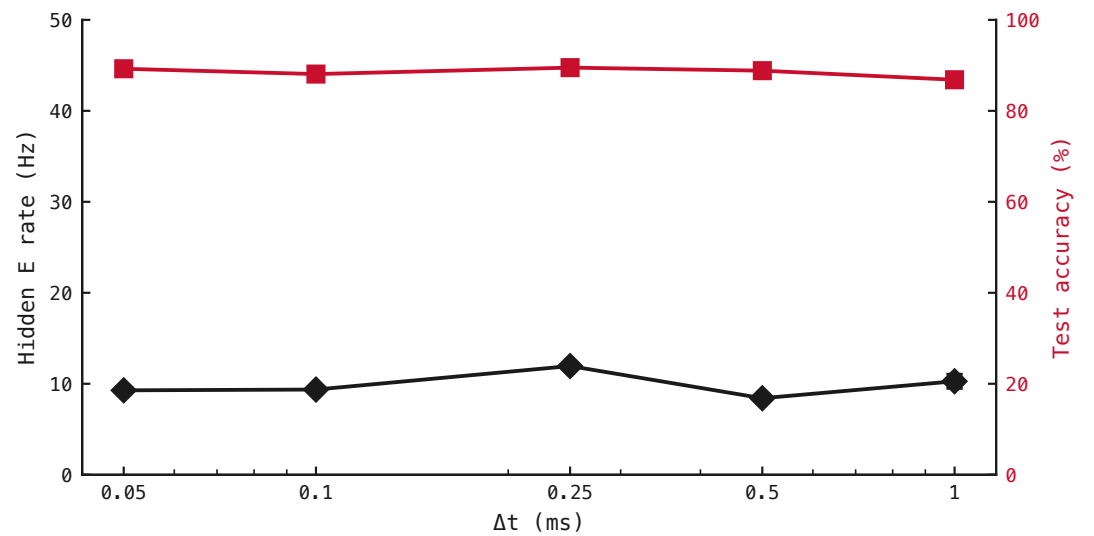
Appendix 0 Figure 9

Gamma gates the rate, not mean inhibition – matched mean I, opposite E response



Appendix 0 Figure 10

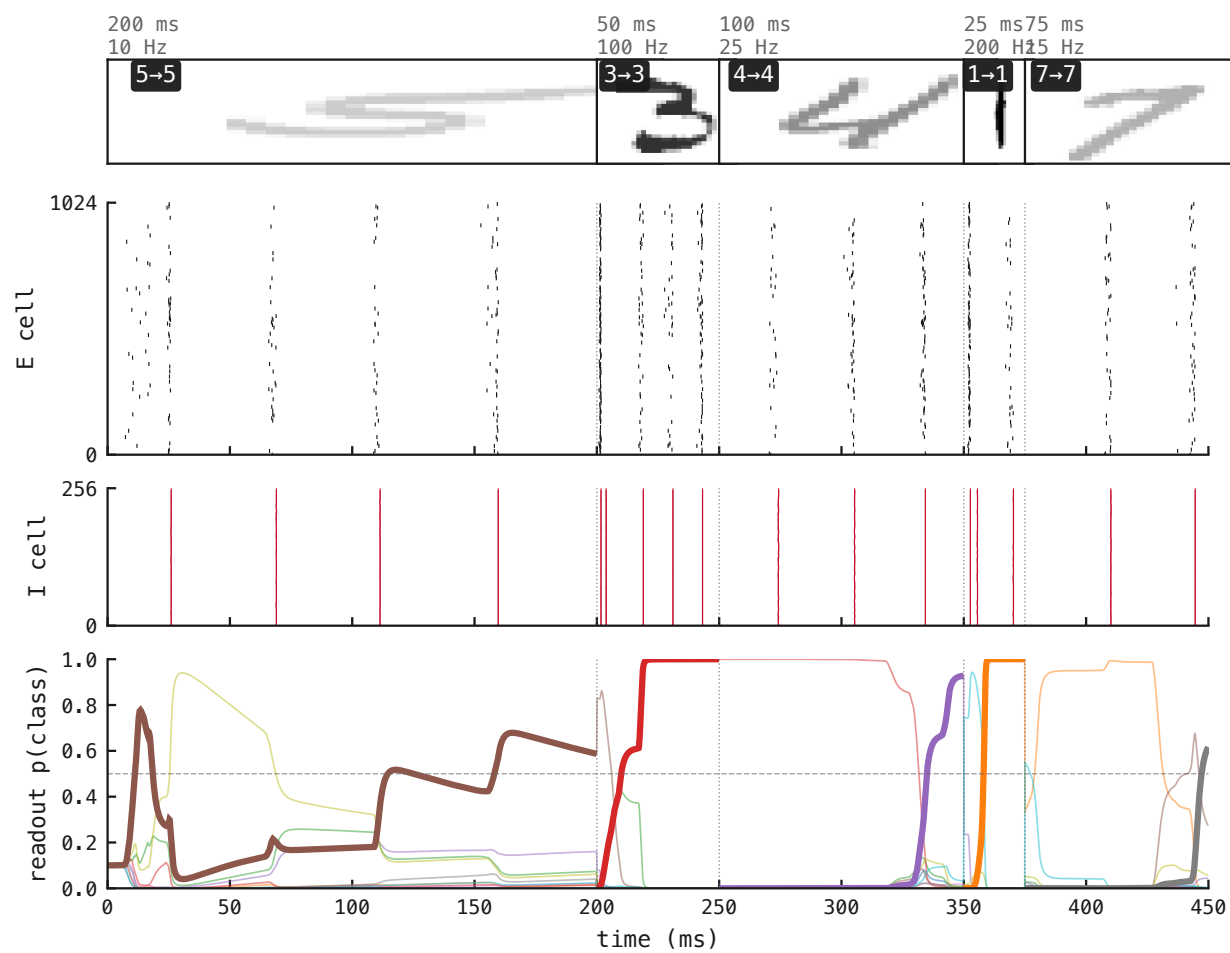
PING Δt audit – post-training E rate and accuracy vs integration timestep



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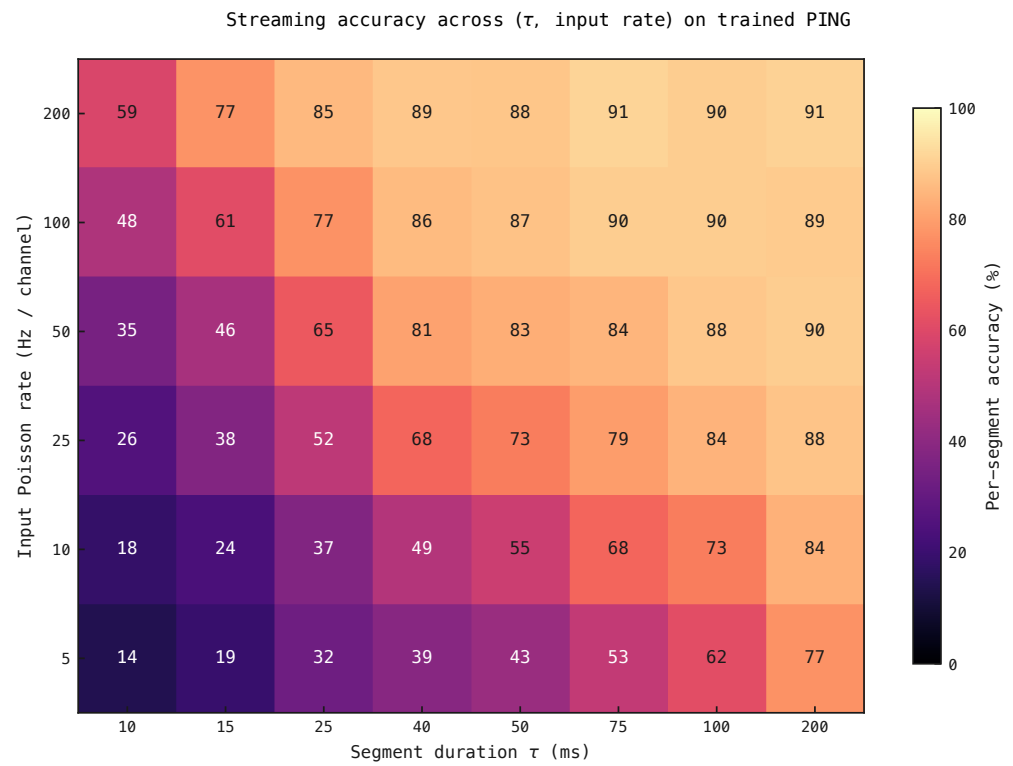
Appendix 0 Figure 11

Streaming digit classification — per-segment (τ , rate) varies, trained PING (no retraining)



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Appendix 0 Figure 12



Appendix 0 Figure 13